

4. Research and Development Documentation:

This chapter presents the three-phase analytical pipeline that was designed to analyse reviews of Amazon apps. The pipeline progresses through three phases: exploratory data analysis (Phase 1), through fake review detection (Phase 2), to sentiment classification (Phase 3) which forms a sequential data processing workflow.

4.1 Phase 1 — Exploratory Data Analysis (EDA)

The first phase focused on understanding the structure, quality, and patterns within the Amazon App Reviews dataset. All analysis was conducted in the Jupyter notebook Code/[EDA.ipynb](#).

Data Loading & Quality Assessment

The raw Amazon App Reviews dataset was loaded and inspected for quality issues. The dataset contained 82,069 reviews with 32 features after initial processing.

The score distribution revealed that the class is highly imbalanced as 45.4% of reviews were 1-star ratings and 29.4% were 5-star ratings, with the remaining 25.2% spread across 2–4 stars. This pattern is common in app review dataset as users tend to leave a review only when they are highly satisfied or dissatisfied.

Table 1: Table of Star rating

Rating	Count	Percentage
1 star	37,285	45.4%
2 stars	8,642	10.5%
3 stars	6,709	8.2%
4 stars	5,334	6.5%
5 stars	24,099	29.4%

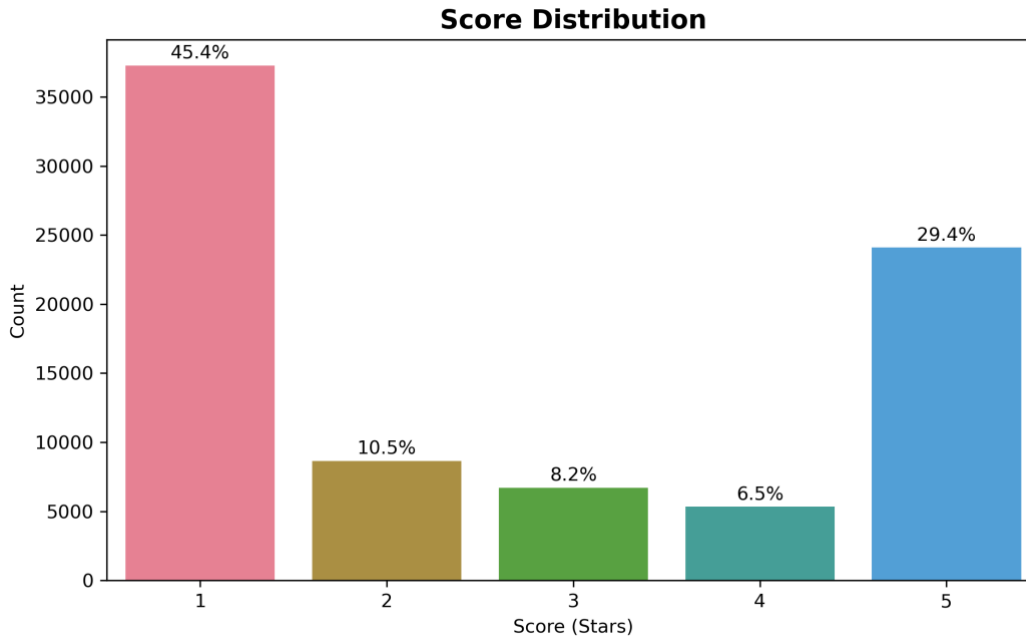


Figure 1: 4.1 Score distribution of Amazon app reviews

Data Cleaning & Preprocessing

For the downstream analysis, several steps were taken to prepare the data. Different temporal features were extracted such as Year, Month, Day_of_week, and hour. The duplicate reviews were removed. The raw review text was cleaned through lowercasing and punctuation normalisation to produce a content_clean column used in subsequent text analysis.

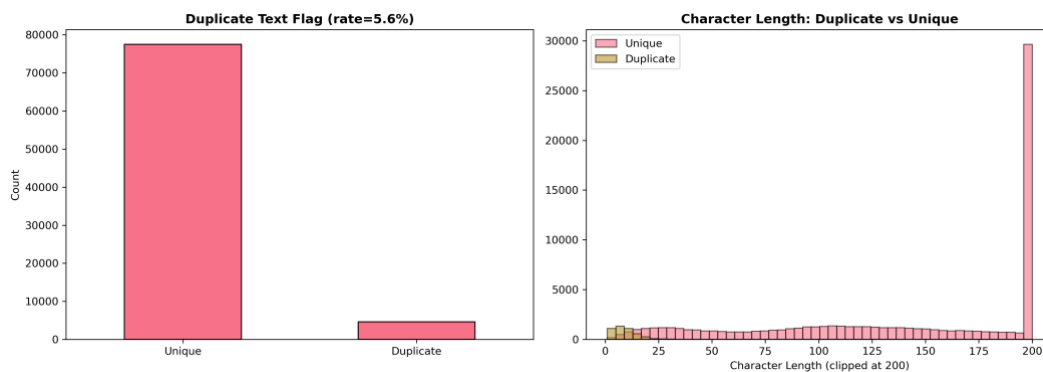


Figure 2: 4.1 Duplicate review analysis

Feature Engineering (26 Numeric Features)

A total of 26 numeric features were engineered from the raw text and metadata to capture various dimensions of review behavior. These features were designed to support both anomaly detection in Phase 2 and general pattern analysis. The features fall into the following categories:

Text length features: word_count, char_len, and avg_word_len are used to capture the substance and complexity of each review.

Punctuation and symbol features: exclamation_count, question_count, caps_ratio, and punctuation_ratio are used to understand the writing style patterns that may indicate spam or manipulation.

Behavioural features: emoji_count, has_url, and repeated_char_flag are used to detect informal or promotional content.

User-level features: is_anonymous_user and reviews_by_username can be used to identify reviewer profiles and prolific posting behaviour.

Lexical features: unique_word_ratio and stopword_ratio assess vocabulary diversity and content substance.

Rating-derived features: score_extremity is used to measure the distance from the neutral rating (3 stars).

Duplicate detection: is_duplicate_text and dup_text_weight flag copy-paste reviews with length-weighted scoring.

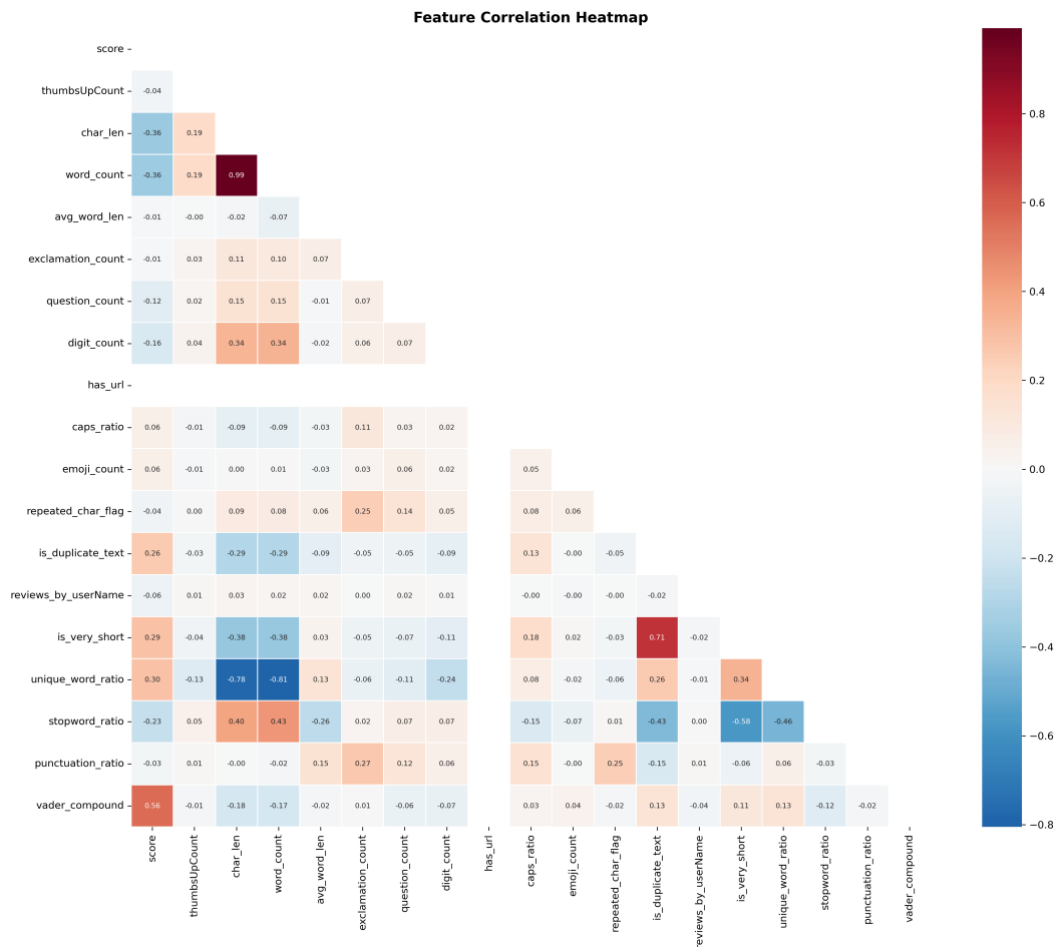


Figure 3: 4.3 Correlation heatmap of engineered features

Key Findings & Visualisations

The exploratory analysis provided several important insights. The word count distribution revealed that most reviews are quite short, with a median of 27 words. However, some reviews exceed 300 words. The thumbs-up distribution was heavily right-skewed. Most reviews received no helpful votes, while a small number received thousands. Temporal analysis showed patterns in when reviews were posted. Certain hours and weekdays had higher activity. The correlation heatmap showed relationships between engineered features. There was a strong correlation between word count and character length, and between score and VADER compound. The VADER sentiment analysis indicated a general agreement between star ratings and sentiment scores based on the lexicon. However, there were notable mismatches, especially with extreme ratings.

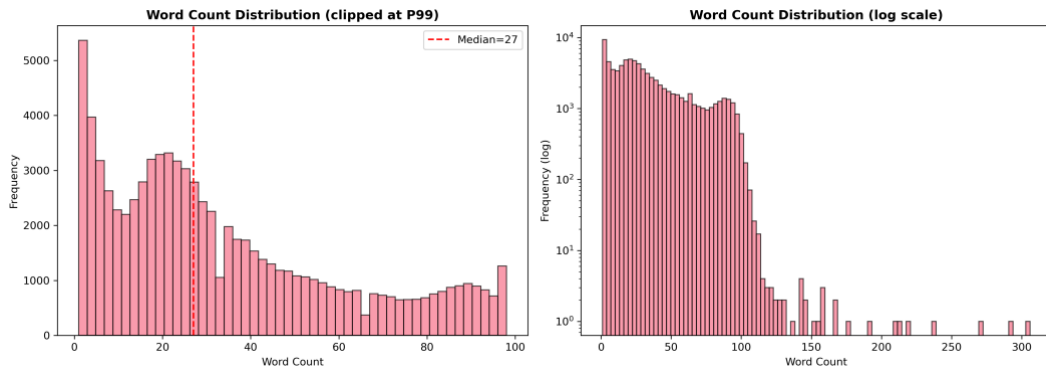


Figure 4: 4.4 Word count distribution

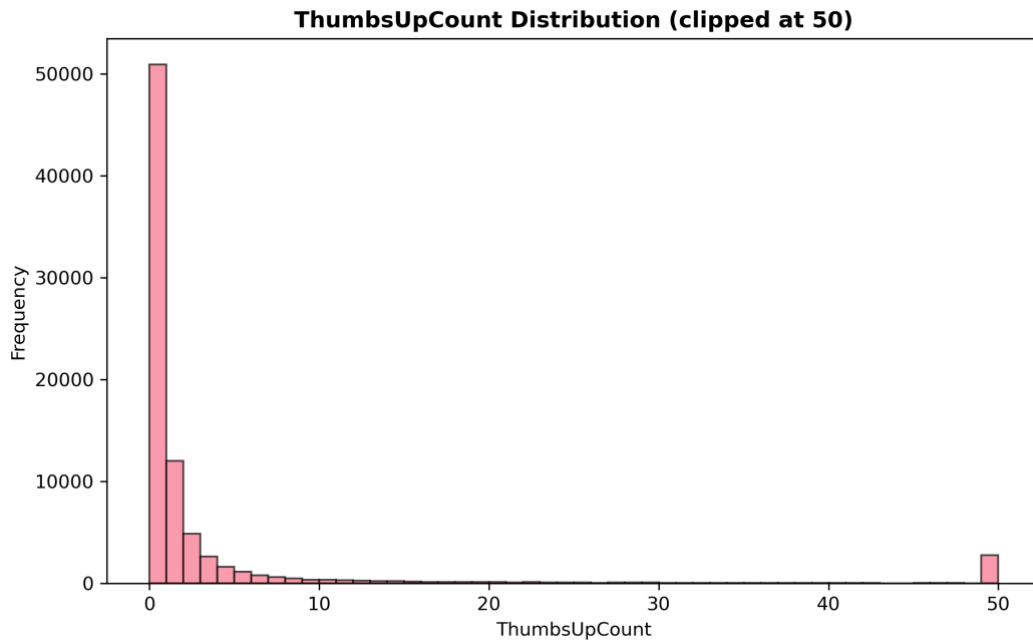


Figure 5: 4.5 Thumps up distribution

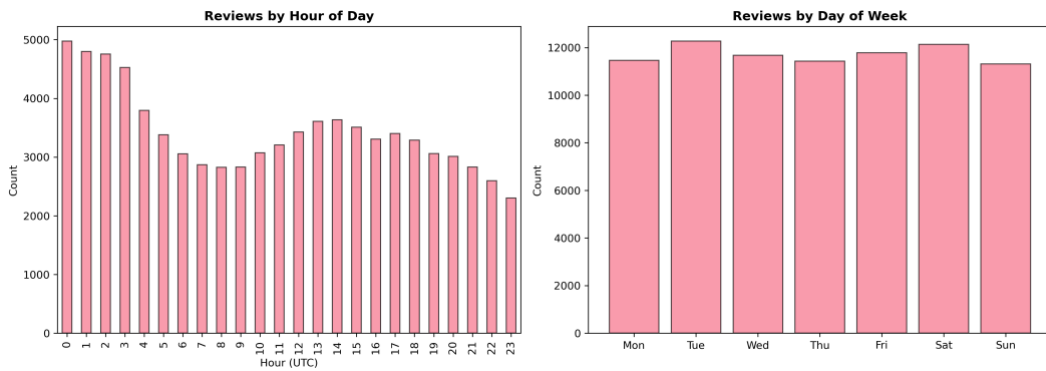


Figure 6: 4.6 Temporal patterns in review posting

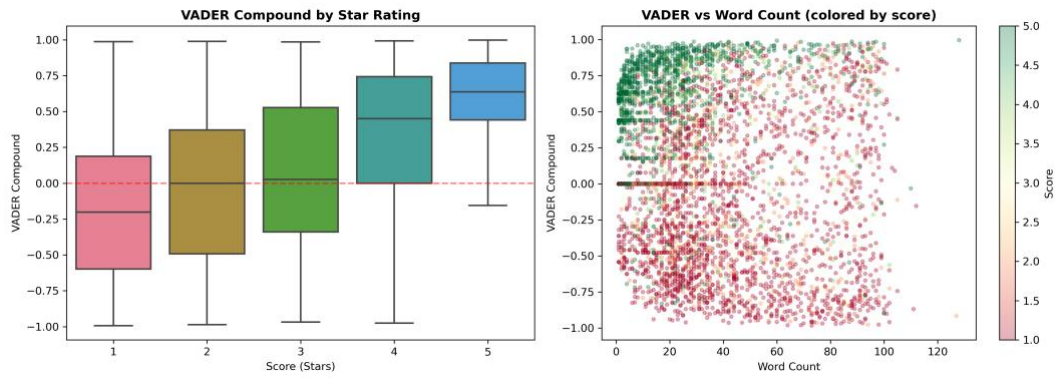


Figure 7: 4.7 VADER Sentiment analysis by rating

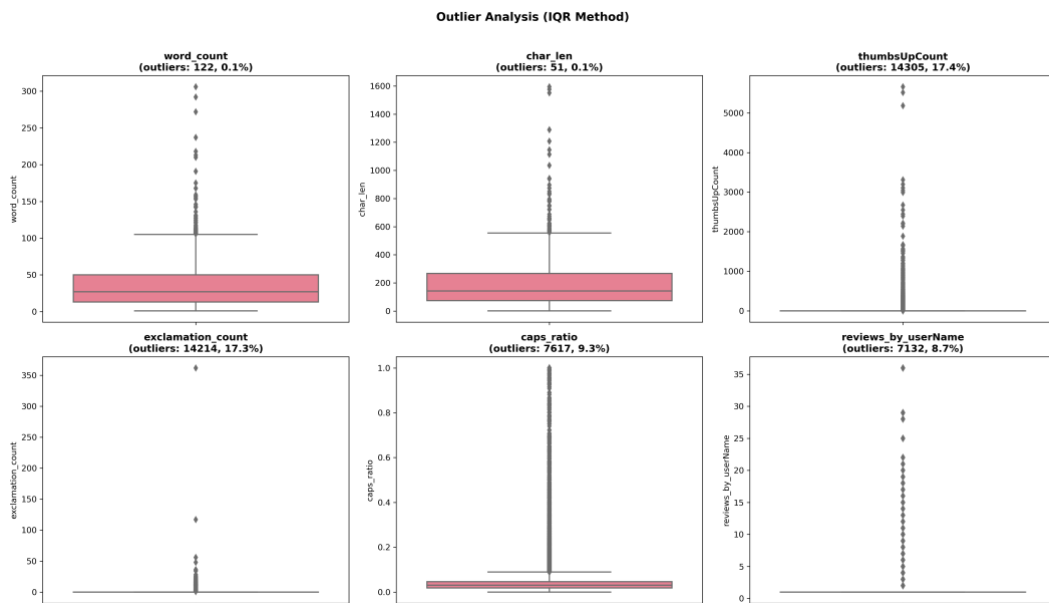


Figure 8: 4.8 Outlier Analysis

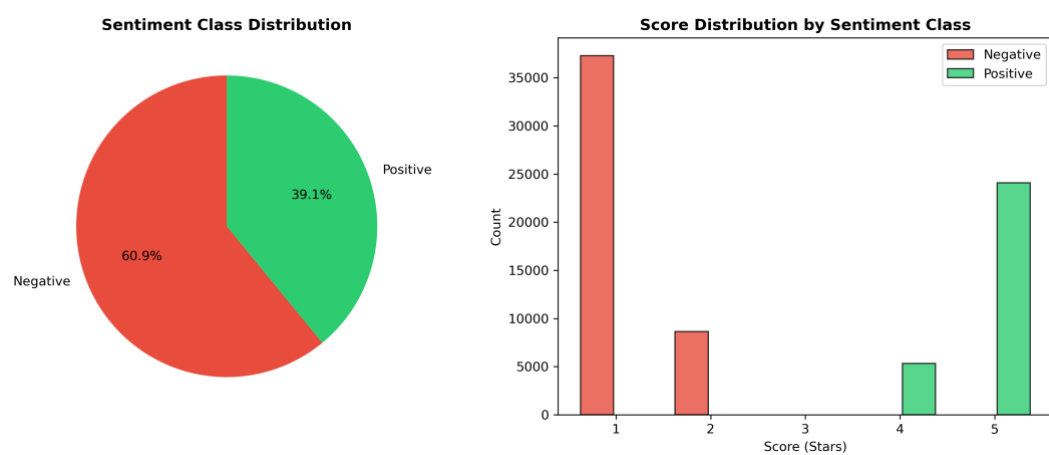


Figure 9: 4.9 Class imbalance analysis

4.1.5 Output Datasets

Two output datasets were produced. The file `reviews_clean_all.csv`, which has 82,069 rows and 32 features, contains all reviews with processed features. This file went to Phase 2 for fake review detection. The file `reviews_clean_full.csv` leaves out neutral (3-star) reviews and includes binary sentiment labels. This serves as another option for direct sentiment analysis.

4.2 Phase 2 — Fake Review Detection

The second phase implemented a hybrid fake review detection system combining rule-based heuristics with unsupervised machine learning. All work was conducted in [“isolationForest.ipynb”](#).

4.2.1 Approach Overview

The detection pipeline was designed with three components: (A) ten rule-based heuristic voters, each targeting a specific pattern of suspicious reviews; (B) an Isolation Forest anomaly detector operating on 21 numeric features; and (C) an ensemble voting system that combines all 11 voters to produce a final fake/genuine classification. This hybrid approach balances interpretability (rules) with the ability to detect complex, non-obvious anomalies (Isolation Forest).

4.2.2 Rule-Based Detection (10 Heuristic Rules)

Ten independent binary rules were defined, each casting one vote when triggered. These rules target common patterns of fake or low-quality reviews identified during EDA:

Table 2: Table of Rule-based detection

Rule	Description	Flagged	%
R1	Generic short review + extreme rating (≤ 3 words, 1 or 5 stars)	6,564	8.00%
R2	Extreme rating + no thumbs up + short (< 10 words)	13,192	16.07%
R3	Excessive capitalisation ($> 50\%$, reviews > 20 chars)	467	0.57%

R4	Weighted duplicate text (dup_text_weight ≥ 0.5)	99	0.12%
R5	Excessive exclamation marks (>5)	984	1.20%
R6	Many emojis (>3)	132	0.16%
R7	VADER-score mismatch (sentiment contradicts rating)	5,171	6.30%
R8	Low lexical diversity (unique_word_ratio <0.4, >10 words)	3	0.00%
R9	Contains URL (promotional/spam)	0	0.00%
R10	Short anonymous review + extreme rating	10	0.01%

In total, 19,646 reviews (23.94%) were flagged by at least one rule. Rule R2 had the highest fire rate at 16.07%, reflecting the large number of short, extreme-rated reviews with no community engagement.

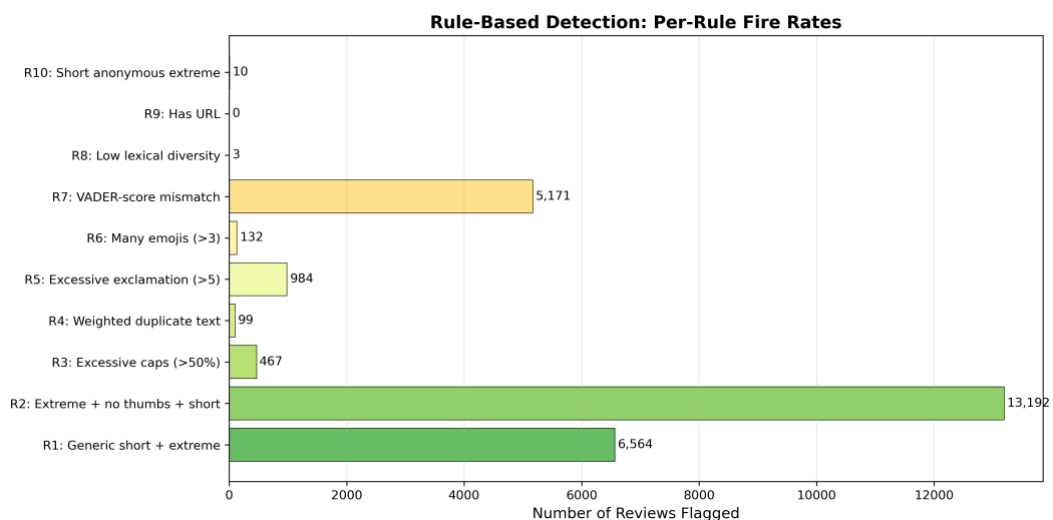


Figure 10: 4.10 Per-rule fire rates

4.2.3 Isolation Forest — Anomaly Detection

Isolation Forest (Liu, Ting, and Zhou, 2008) was selected as the unsupervised anomaly detection algorithm. This method functions by randomly selecting a feature and a split value to recursively partition the data. Anomalous points are isolated with fewer splits, resulting in shorter mean path lengths and lower anomaly scores. The algorithm was selected due to three main advantages: it is tree-based and scale-invariant, eliminating the need for feature normalization; it has $O(n \log n)$ processing complexity and completed training within seconds on 82,069 reviews; and it provides interpretable anomaly scores and feature significance rankings. The model was configured with `n_estimators=200`, increased from the default of 100 to increase stability, `contamination=0.10`, and `random_state=42`. Twenty-one numeric features from Phase 1 served as input. The model identified 8,207 anomalies, representing 10.00% of all reviews.

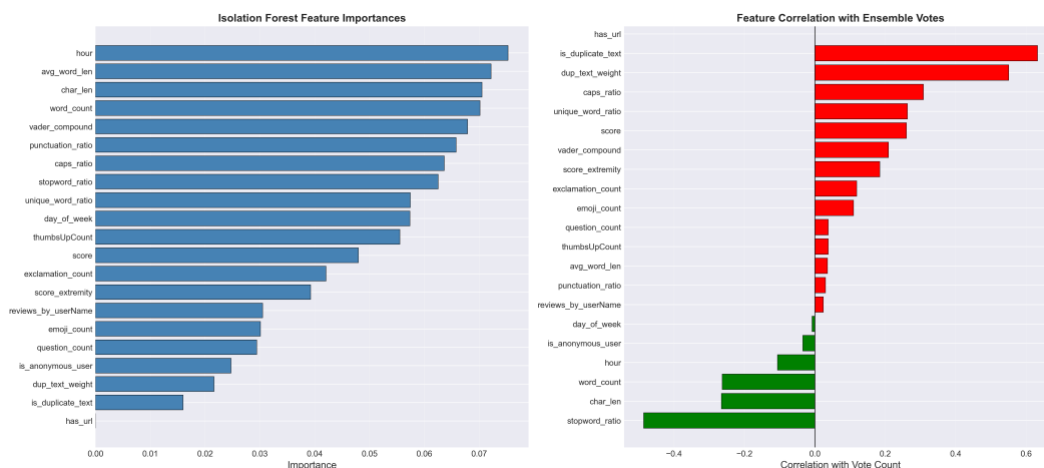


Figure 11: 4.11 Isolation forest feature importance

4.2.4 Contamination Sensitivity Analysis

A sensitivity analysis was performed to justify the contamination parameter selection. Three contamination rates were tested:

Table 3: Table of Contamination sensitivity analysis

Contamination Rate	Anomalies Detected	Percentage
0.05	~4,103	5.00%
0.10	8,207	10.00%
0.15	~12,310	15.00%

The rate of 0.10 was selected as a balanced choice between precision (avoiding false positives) and coverage (catching genuine anomalies). A rate of 0.05 was considered too conservative, while 0.15 risked removing legitimate reviews.

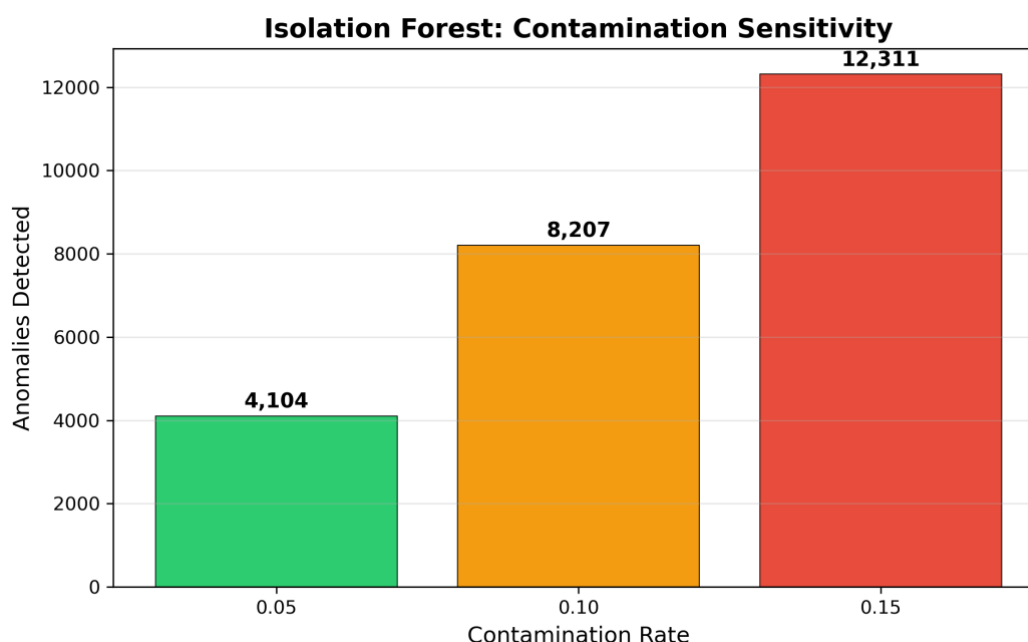


Figure 12: 4.12 Contamination sensitivity analysis

4.2.5 Ensemble Voting & Threshold Selection

All 11 voters (10 rules + Isolation Forest) were combined into an ensemble system. Each review received a total vote count (0–11), and three threshold levels were evaluated:

Table 4: Table of Ensemble voting and threshold selection

Threshold	Votes Required	Reviews Flagged	Percentage
Conservative	≥ 4	50	0.06%
Moderate	≥ 3	2,873	3.50%
Aggressive	≥ 2	8,218	10.01%

The moderate threshold (≥ 3 votes) was selected for the final pipeline which led to the removal of about 2,873 suspicious reviews and the remaining data were used in phase 3. This removed 2,873 suspicious reviews, producing a clean dataset of 79,196 reviews for Phase 3.

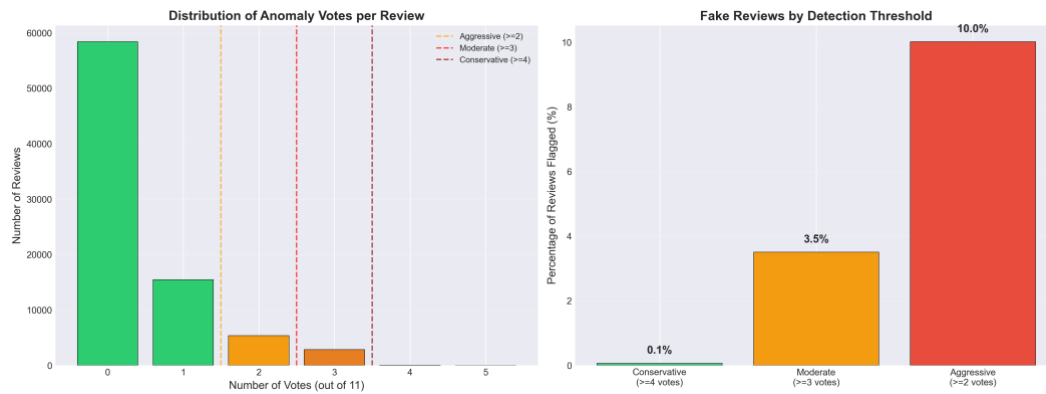


Figure 13: 4.13 Ensemble vote distribution

4.2.6 Validation & Results

The detection results were validated through several analyses. An inter-method agreement heatmap showed the correlation between each voter, confirming that rules and the Isolation Forest captured complementary patterns. A PCA visualisation projected reviews into two dimensions, showing visible separation between flagged and clean clusters. Feature importance analysis revealed which features contributed most to Isolation Forest decisions. A comparison of fake versus real review characteristics confirmed that flagged reviews exhibited shorter lengths, more extreme ratings, and lower lexical diversity than genuine reviews.

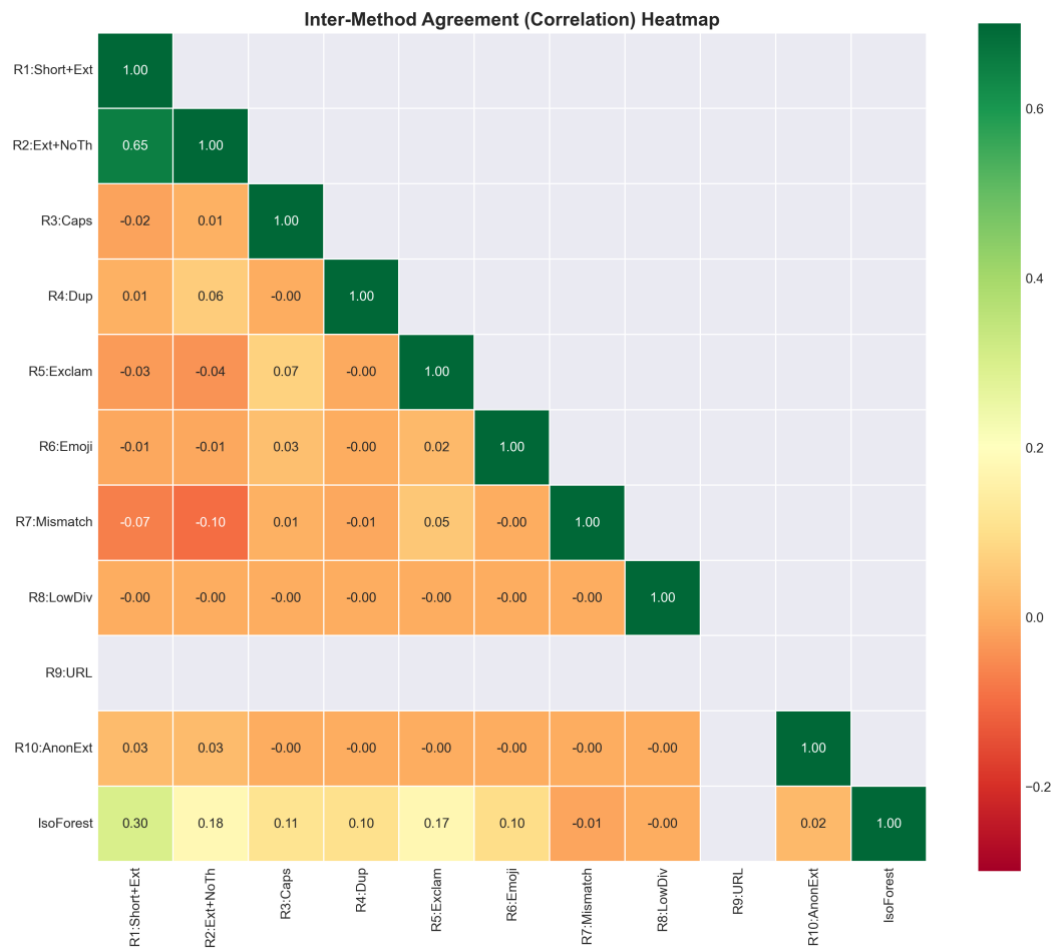


Figure 14: 4.14 Inter-method agreement heatmap

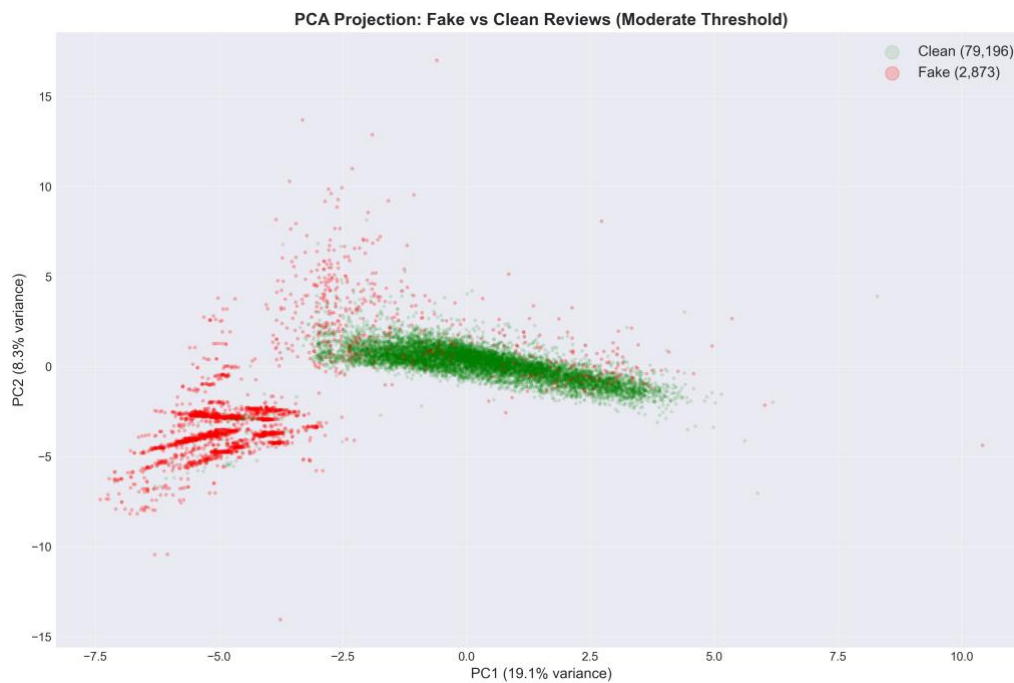


Figure 15: 4.15 PCA - Fake vs Clean Reviews

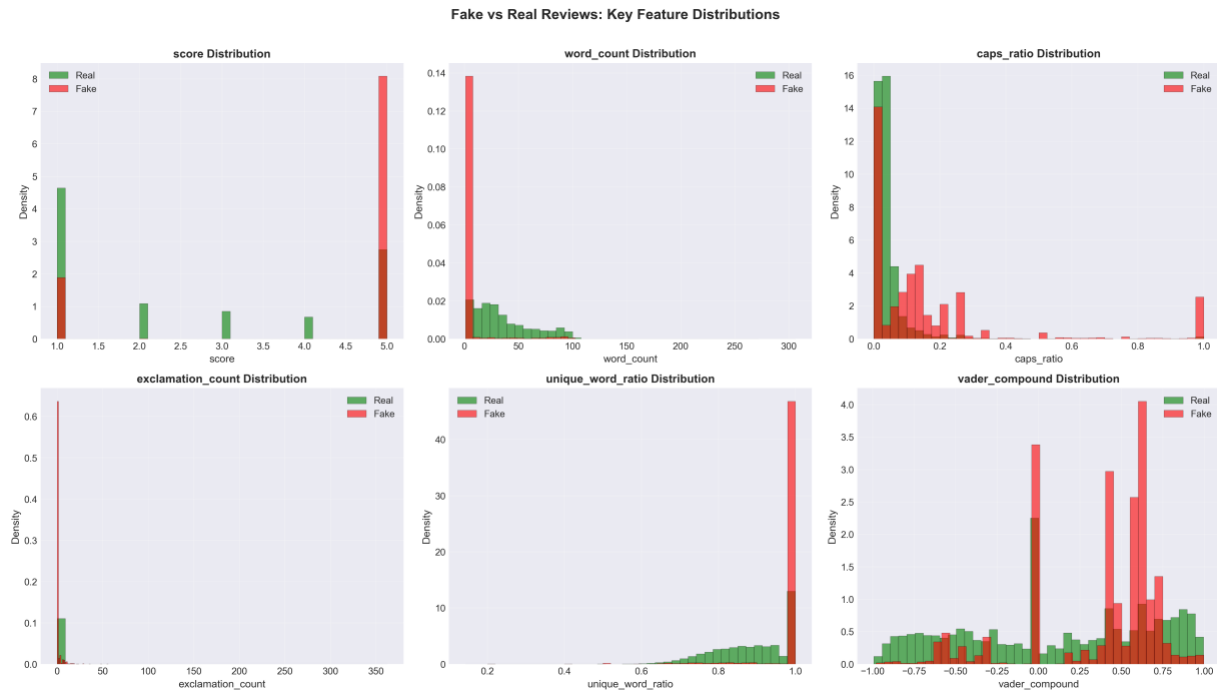


Figure 16: 4.16 Fake vs Real review characteristics

4.3 Phase 3 — Sentiment Analysis

The third phase built and evaluated sentiment classification models on the cleaned dataset (79,196 reviews with fake reviews removed). All work was conducted in Code/[sentimentAnalysis_2models.ipynb](#).

4.3.1 Data Preparation & TF-IDF Vectorisation

Three labels were defined as Negative (1-2 stars), Positive (4-5 stars) and Neutral as (3-star) which was excluded resulting in 72,485 samples (45,379 Negative; 27,106 Positive). 57,988 training and 14,497 test samples were produced by a stratified 80/20 split. Using TF-IDF (max_features=10,000, ngram_range= (1,2)) was used to vectorize the text which enables the model to capture key phrases like “not good” and “very bad”.

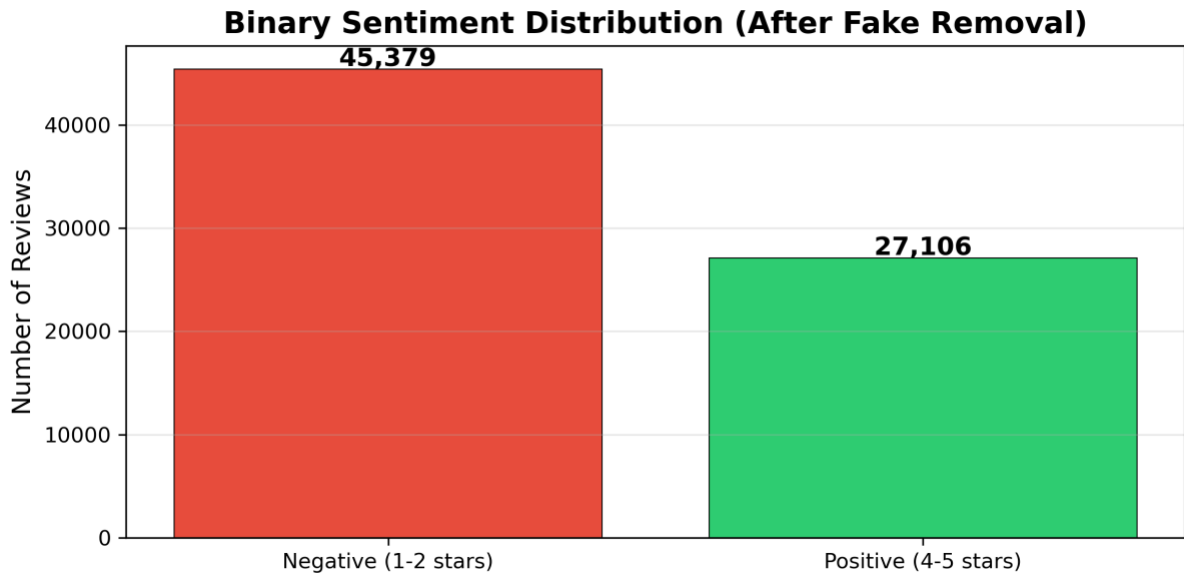


Figure 17: 4.17 Sentiment distribution

4.3.2 Model Training (Logistic Regression & Naive Bayes)

Two classifiers were trained and evaluated:

Logistic Regression: L2-regularised with balanced class weights to handle the class imbalance between negative and positive reviews.

Multinomial Naive Bayes: Configured with Laplace smoothing ($\alpha=0.1$) to handle zero-frequency terms in the TF-IDF matrix.

4.3.3 Model Comparison & Evaluation

Both models were evaluated on the held-out test set and through 5-fold stratified cross-validation. The results are summarized below:

Table 5: Table of Model comparison

Metric	Logistic Regression	Multinomial NB
Accuracy	0.9228	0.9049
F1 (Weighted)	0.9227	0.9027
F1 (Negative)	0.9386	0.9279
F1 (Positive)	0.8961	0.8604
Precision	0.9227	0.9097
Recall	0.9228	0.9049
AUC-ROC	0.9678	0.9561
CV Mean F1 (5-fold)	0.9214 $\pm$ 0.0016	0.9063 $\pm$ 0.0020

Logistic Regression was selected as the best model with F1 score of 0.9227 as it outperformed Multinomial Naive Bayes. The model was validated using confusion matrices and ROC curves with an AUC of 0.9678. The strong generalization was shown by the 5-fold cross validation with a small train test gap of 0.0179.

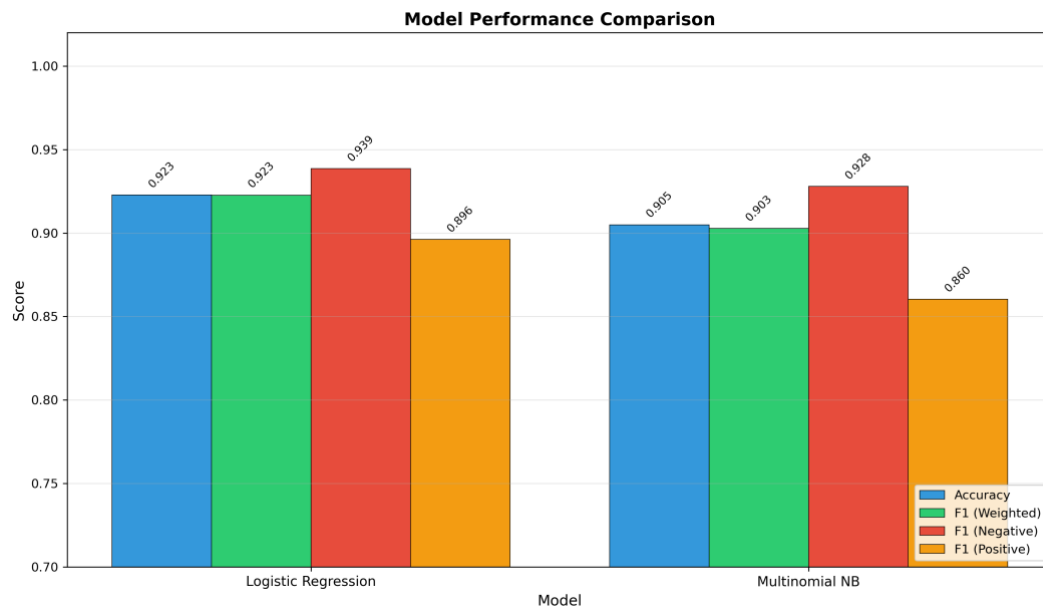


Figure 18: 4.18 Model comparison

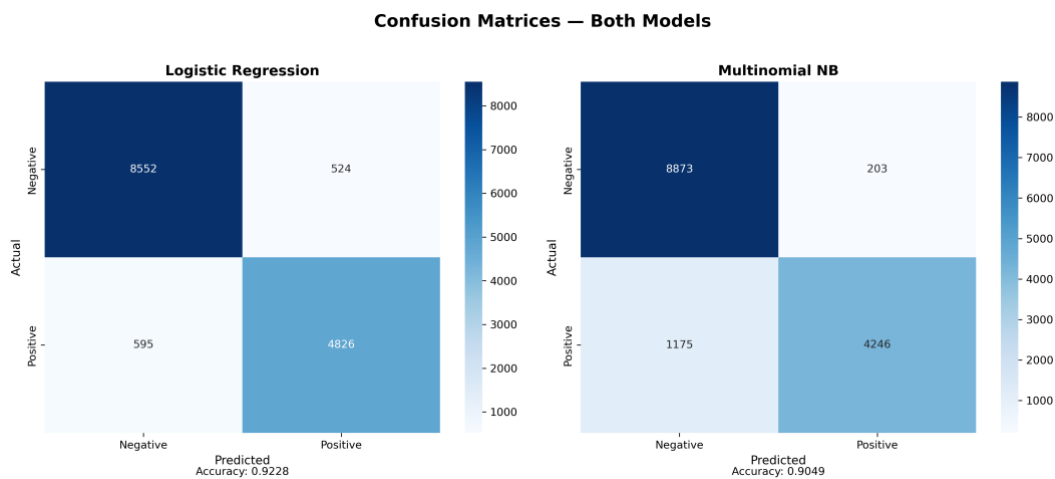


Figure 19: 4.18 Confusion matrices

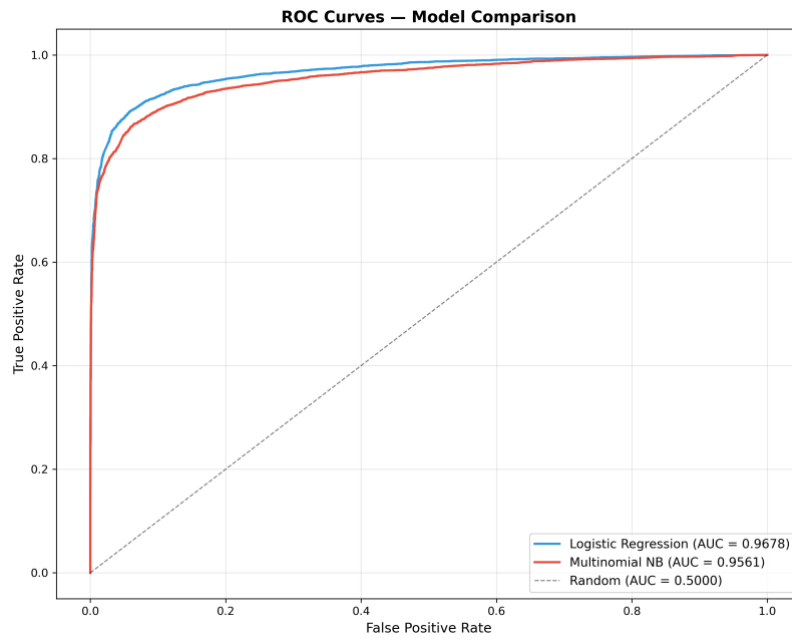


Figure 20: 4.20 ROC Curves

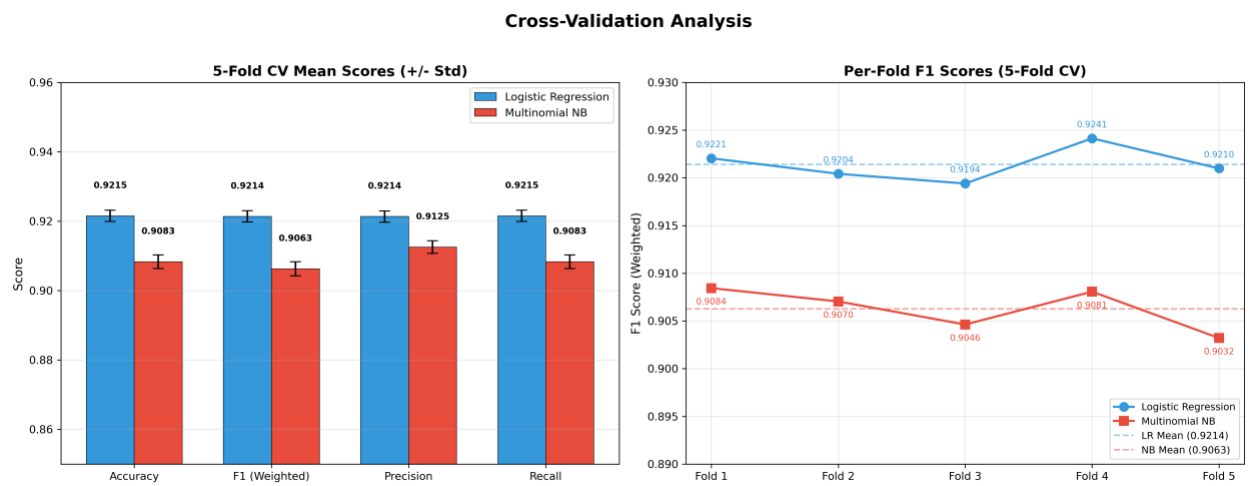


Figure 21: 4.21 5-fold cross-validation results

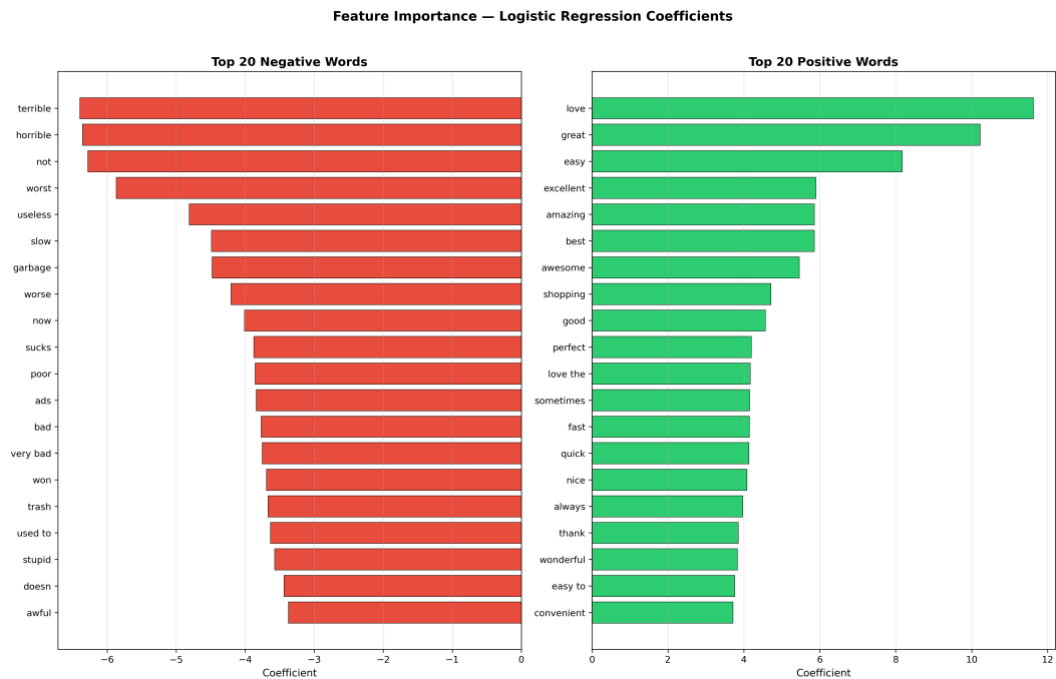


Figure 22: 4.22 Top words per sentiment class

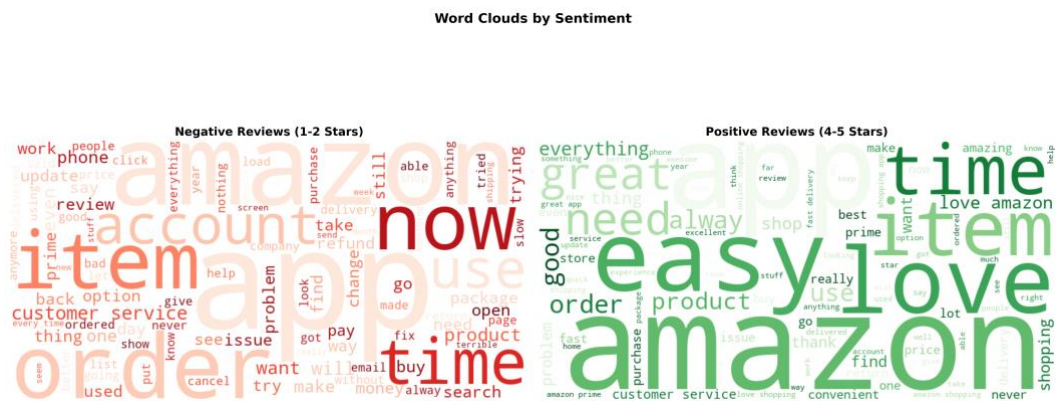


Figure 23: 4.23 Word clouds - Positive vs Negative

4.3.4 Impact Analysis — Effect of Fake Review Removal

Two datasets were used to train the logistic regression model in which one was the filtered dataset (79,196 reviews) and another one was the unfiltered dataset (82,069 reviews). The model which was trained on unfiltered data achieved a slightly better F1 score (0.9263 vs. 0.9227, +0.39%), likely because fake reviews are often short and are easier to classify, which inflate the performance. However, the filtered model provides more reliable predictions on genuine reviews, which is the main goal of the pipeline.

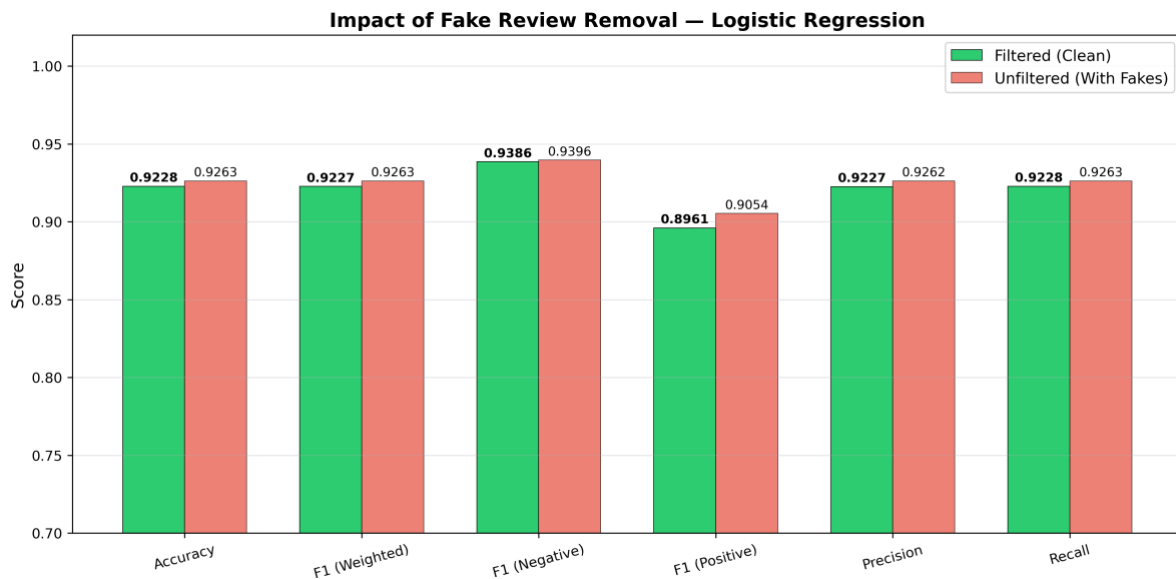


Figure 24: 4.24 Impact of fake reviews removal on sentiment accuracy

4.4 Phase 4 — Interactive Dashboard

A web dashboard was built to ensure even a non-technical user could understand the results easily. The dashboard was built using Streamlit along with Plotly. It combined all the result from earlier phases (data, fake review detection, sentiment results and model performance) into a unified dashboard.

4.4.1 Architecture

The multipage dashboard is designed which caches large dataset to improve the performance. A common stylesheet is being used to ensure consistent design. The model is trained only once at startup, and it is cached to enable fast inference.

4.4.2 Page 1 — Project Overview

The first page is landing page called Project overview which summarizes the pipeline with KPIs tiles showing different information such as total reviews, fake reviews at different threshold such as moderate (2,873), also showing the best model accuracy and many other such important information. It also has some interactive rating distribution chart and a comparison table for both models.

4.4.3 Page 2 — Fake Review Detection

The second page is mainly used to show the phase 2 results. Different tiles are displayed showing the KPIs, rule-based diagnostics, feature comparisons, Isolation Forest insights, PCA visualisation, agreement heatmaps and sensitivity analysis.

4.4.4 Page 3 — Sentiment Analysis

This page is mainly used to display information about the phase 3. Different information is laid out in KPI tiles which summarizes different information. Also, different metrics tables and visualisations are shown such as model comparisons, confusion matrices, ROC curves, cross-validation results, sentiment distribution and key word analysis.

4.4.5 Page 4 — Live Prediction

This page is used to show prediction for the user review. The user can submit a single review or a csv file containing bunch of reviews to predict the sentiment and check if the review is genuine.